Q1 Event Communication Security

a) Reconstruct the encrypted message using a Shift of 4 position.

Give msg: MEET AT NOON

Encryption : Caesar Cipher

Shift : 4 positions Forward.

M - Q
E - I
E - I
T - X
A - E
T - X
N - R
O - S
O - S
N - R

Encrypted msg: QIIXEXRSSR

Spaces remain unchanged.

b) The authorized member uses the same Shift of 4 to decrypt the Cipher text

Step:

* Take each encrypted letter
* Repeat For every letter

For Ex:

Q → M
I → E
X → T

Therefore the decrypted message in MEET AT NOON

2)

a) Find the original msg

Give cipher txt:

GSRH RH Z HVXIVG

The msg uses the Atbash cipher, a substitution cipher in which the alphabet is reversed.

The Substitution pairs are:

A ↔ Z, B ↔ Y, C ↔ X, D ↔ W and soon.

Decrypt each word:

CIPHER TEXT	PLAIN TEXT
GSRH	THIS
RH	IS
Z	A
HVXIVG	SECRET

b) Justify your answer:

- The Atbash cipher replaces each letter with its Reverese -alphabet counter part.
- The first word, GSRH becomes THIS
- The decoded sentence is meaningful English, Confirming that. result.

3.

a) Encrypt HELLO using the keyboard KEY

Plain text : HELLO

Keyword : KEY

The Vigenere cipher use a repeating keyword to determine the Shift For Each other plain txt letter.

H E L L O

K E Y K E

Step 2: Convert letters into numerical values

use A=0, B=1, C=2, ... Z=25

Letter : H E L L O

Plaintext Value : 7 4 11 11 14

keyword letter : K E Y K E

keyword value : 10 4 24 10 4

$C = P + K \mod 26$

$H + K = 7+10 \mod 26 = 17 \rightarrow R$

$E + E = 4+4 \mod 26 = 8 \rightarrow I$

$L + Y = 11+24 \mod 26 = 9 \rightarrow J$

$L + K = 11+10 \mod 26 = 21 \rightarrow V$

$D + E = 4+4 \mod 26 = 18 \rightarrow S$

RIJVS

b) Each keyword letter determines the Shift applied to the corresponding plain text letter.

- The keyword KEY is repeated to match the length of HELLO
- Diff. keyword letters produce diff. Shifts
- The receiver must know the same keyword to decrypt the message

4)

a) How can Investigators detect hidden data

Steganography is a technique used to hide Confidential information inside an ordinary file such as an image, without making the hidden information obvious

Investigators can use the following methods:

Visual Inspection:

Examine the Image For unsual colors. Patterns or distortions

b)

1) Encryption before embedding:

Encryption the confidential msg using a Strong encryption algorithm before hiding it inside the image

- Even if Someone Extract the hidden msg, they Cannot easily understand it without the decryption.

5)

a) Find the original message by reversing the Encryption Process

Given Cipher txt:

WECRLTEERDSOEEFEADCAIVDEN

- This question refers to a transposition cipher with three levels of arrangement.

The most probable original msg is

WEARDISOVERDDFLEEATONCE

For the Plaintxt:

~~WE C R C T E E~~.

WE ARE DISCOVERED FLEE AT ONCE,

Rail:

WECRLTEERDSOEEFEAOCAIVDEN

Chiper text

No character Subsitiution:

The original letters remain unchanged; only their positions are rearranged.

Encryption